

**Toka Priyanka**

[tokapriyanka@gmail.com](mailto:tokapriyanka@gmail.com)

**AI/ML intern**

**Role: Computer Vision**

# **3D Virtual Try-On Research & Prototype Report**

## **1. Introduction**

3D Virtual Try-On technology creates immersive shopping experiences by digitally fitting garments onto realistic 3D avatars. Unlike 2D try-on systems, 3D systems use body meshes, avatars, cloth simulation, and motion tracking to generate realistic garment behavior and fitting visualization.

The objective of this research phase is to study and prototype the components required for developing a scalable and realistic 3D virtual try-on system.

The system focuses on avatar generation, body mesh fitting, cloth simulation, movement tracking, rendering quality, and user measurement integration. The uploaded research notes already identify avatars, meshes, cloth simulation and motion tracking as the major components.

## **2. Objectives**

The main objectives of this research work are:

- Study 3D avatar generation techniques
- Explore body mesh fitting methods
- Improve cloth simulation realism
- Analyze real-time movement tracking
- Investigate rendering optimization methods
- Integrate user measurements
- Design prototype workflow architecture

## **3. Proposed 3D Try-On Workflow**

Proposed system architecture:

User Image / Camera Input



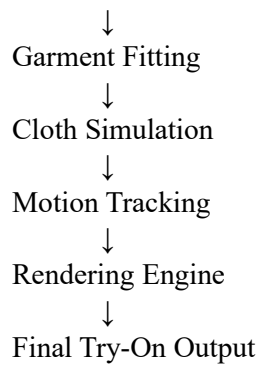
Body Detection



Avatar Generation



3D Mesh Creation



#### **4. 3D Avatar Generation**

##### **Objective**

Generate realistic virtual avatars representing user body structures.

3D avatars act as the digital representation of users for garment fitting and visualization.

##### **SMPL Model**

SMPL (Skinned Multi-Person Linear Model) generates parametric human body meshes.

Features:

- Adjustable body proportions
- Pose parameter support
- Realistic human structure
- Mesh deformation support

Benefits:

- Better personalization
- Scalable body representation
- Supports garment fitting

##### **PIFuHD**

PIFuHD converts human images into detailed 3D body representations.

Capabilities:

- High-resolution reconstruction
- Surface estimation
- Detailed body extraction
- Realistic avatar generation

The research notes recommend SMPL and PIFuHD for body mesh creation.

## **5. Body Mesh Fitting**

### **Objective**

Fit garments accurately onto avatar surfaces.

Mesh fitting aligns garments according to:

- Body size
- Body shape
- User posture
- Surface geometry

### **Mesh Deformation**

Mesh deformation modifies garment geometry according to body structure.

Operations:

- Vertex adjustment
- Surface mapping
- Cloth stretching
- Shape alignment

Benefits:

- Improved fitting
- Better cloth contact
- Realistic appearance

Research notes indicate mesh deformation improves alignment between body and garments.

## **6. Cloth Simulation**

### **Objective**

Simulate realistic cloth behavior.

Cloth simulation improves garment realism by reproducing:

- Folding
- Gravity effect
- Stretching
- Wrinkles

- Collision behavior

### **PhysX Engine**

PhysX performs physical simulation.

Features:

- Cloth physics
- Collision handling
- Dynamic movement
- Realistic response

### **Unreal Engine Cloth System**

Unreal Engine improves visual simulation.

Capabilities:

- High-quality rendering
- Dynamic cloth response
- Physics integration
- Real-time visualization

The uploaded notes identify PhysX and Unreal Engine as preferred simulation environments.

## **7. Avatar Movement Tracking**

### **Objective**

Synchronize avatar movement with users.

Movement tracking enables avatars to imitate human actions.

### **MediaPipe Integration**

MediaPipe extracts body landmarks.

Detected points:

- Shoulder
- Elbow
- Wrist
- Hip
- Knee

- Ankle

Advantages:

- Fast inference
- Real-time tracking
- Lightweight execution

### **Motion Tracking Pipeline**

Camera Input



Pose Detection



Landmark Extraction



Avatar Motion Mapping



Animation Update

Research notes indicate MediaPipe supports real-time avatar movement.

## **8. Rendering Optimization**

### **Objective**

Improve visualization quality and responsiveness.

Rendering determines visual realism of the final output.

Requirements:

- High FPS
- Low latency
- Realistic lighting
- Cloth texture preservation

### **GPU Rendering**

GPU acceleration improves:

- Rendering speed
- Visualization quality
- User interaction
- Real-time updates

Benefits:

- Reduced lag
- Better realism
- Smooth experience

GPU rendering support is highlighted in the research notes.

## 9. User Measurement Integration

### Objective

Improve personalization accuracy.

Measurements used:

- Height
- Shoulder width
- Chest size
- Waist size
- Hip size
- Arm length

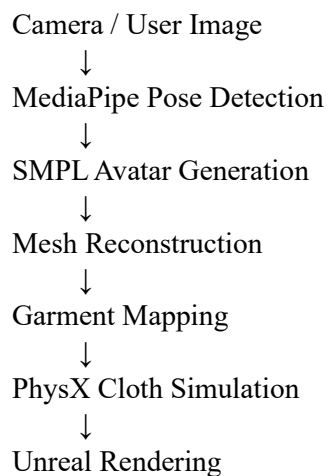
Applications:

- Garment sizing
- Body scaling
- Fit prediction
- Personalized recommendation

Research notes emphasize measurement integration for precision fitting.

## 10. Prototype Architecture

Proposed prototype:



↓  
3D Try-On Output

## **11. Challenges**

Major challenges observed:

### **Rendering Challenges**

- High computational load
- GPU dependency
- Latency

### **Cloth Challenges**

- Collision detection
- Cloth penetration
- Wrinkle generation

### **Avatar Challenges**

- Body proportion mismatch
- Shape variation
- Realistic movement

### **System Challenges**

- Real-time constraints
- Memory usage
- Scalability

Research notes mention rendering and cloth collision as major issues.

## **12. Future Improvements**

Planned enhancements:

- Better avatar realism
- Improved cloth physics
- Real-time optimization
- Dynamic measurement system
- Advanced mesh fitting

- Full body tracking
- Multi-angle rendering

### **13. Conclusion**

The research and prototype study explored major components required for developing a 3D virtual try-on system including avatar generation, body mesh fitting, cloth simulation, movement tracking, rendering optimization, and user measurement integration.

The study establishes the foundation for future implementation of immersive real-time virtual fitting systems with improved realism and personalization.